

CompTIA.

A+ Core 1

Exam 220-1201

A+ 220 1201

Module 5



Comparing Local Networking Hardware

Learning Outcomes

Prepare for A+ Core 1 by:

- Comparing network types
- Comparing networking hardware
- Explaining network cable types
- Comparing wireless networking types

Lesson 5.1

Network Types

LANs and WANs

Local Area Network (LAN)

- Single geographic location

Wide Area Network (WAN)

- Large geographic location

Wireless Local Area Network
(WLAN)

- IEEE 802.11

Metropolitan Area Network
(MAN)

Personal Area Network (PAN)

Storage Area Network (SAN)

SOHO and Enterprise Networks

- Small Office/Home Office (SOHO)
 - LAN with single network appliance for connectivity
- Enterprise
 - LAN with separate devices for network connectivity

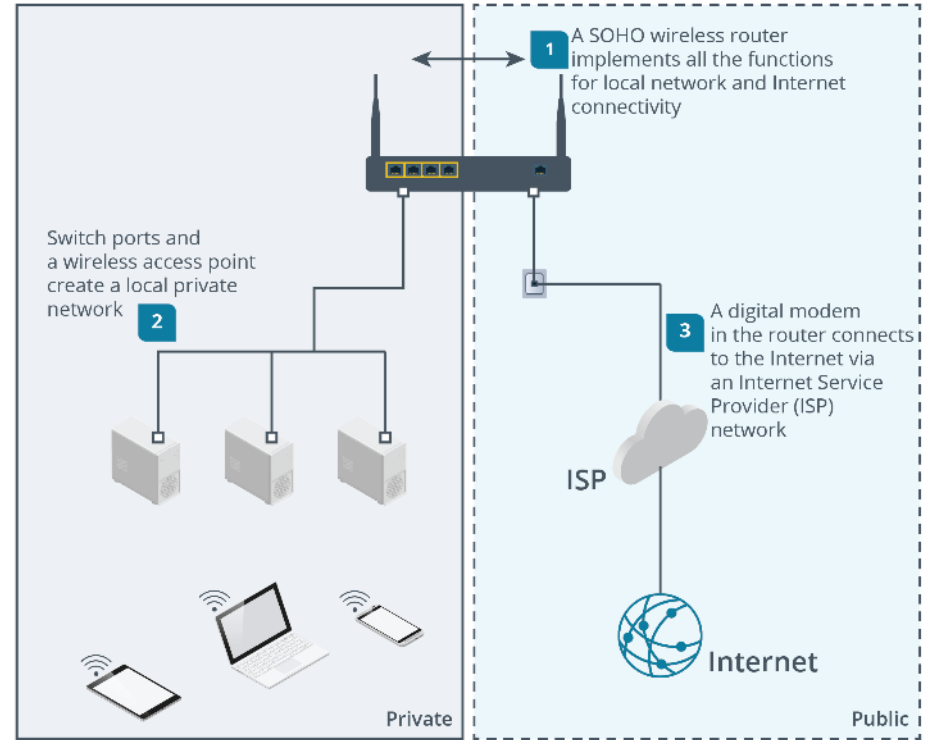
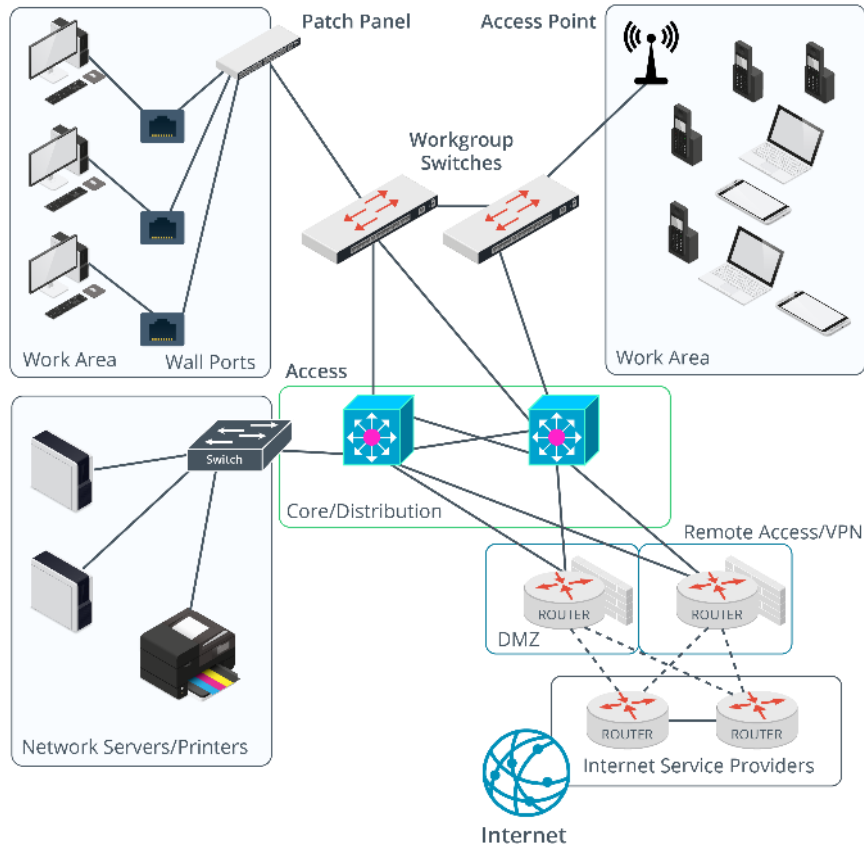


Image © 123RF.com.

SOHO and Enterprise Networks



- Positioning of network components

Datacenters



Site housing server resources



Dedicated network, power, climate-controlled, with secure access controls to protect the environment

Lesson Review: Network Types



LANS AND WANS



SOHO AND ENTERPRISE
NETWORKS



DATACENTERS

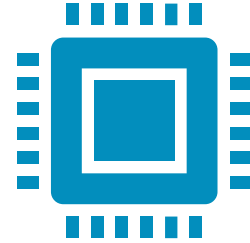
Lesson 5.2

Networking Hardware

Network Interface Cards



Ethernet, fiber, and wireless



MAC address = hardware address

48-Bit, hexadecimal digits

- 24-bit OUI
- 24-bit NIC specific

Network Interface Cards

The screenshot displays the Wireshark network protocol analyzer interface. The title bar reads "Capturing from WiFi". The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for starting/stopping capture, saving, opening, and other standard functions. A display filter bar at the top shows "Apply a display filter ... <Ctrl-/>" with a search icon and a plus sign for expressions.

The packet list pane shows two captured packets:

No.	Time	Source	Destination	Protocol	Length	Info
3	2.3...	IntelCor_50:38:04	Broadcast	ARP	42	Who has 192.168.1.247? Tell 192.168.1.73
4	2.9...	IntelCor_50:38:04	Broadcast	ARP	42	Who has 192.168.1.247? Tell 192.168.1.73

The packet details pane for the selected packet (Frame 3) shows:

- > Frame 3: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
- ▼ Ethernet II, Src: IntelCor_50:38:04 (ac:72:89:50:38:04), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 - ▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
Address: Broadcast (ff:ff:ff:ff:ff:ff)
.... ..1. = LG bit: Locally administered address (this is NOT the factory default)
.... ..1. = IG bit: Group address (multicast/broadcast)
 - ▼ Source: IntelCor_50:38:04 (ac:72:89:50:38:04)
Address: IntelCor_50:38:04 (ac:72:89:50:38:04)
.... ..0. = LG bit: Globally unique address (factory default)
.... ..0. = IG bit: Individual address (unicast)
 - Type: ARP (0x0806)
- > Address Resolution Protocol (request)

The packet bytes pane shows the raw data in hexadecimal and ASCII:

Offset	Hex	ASCII
0000	11111111 11111111 11111111 11111111 11111111 11111111 10101100 01110010	r
0008	10001001 01010000 00111000 00000100 00001000 00000110 00000000 00000001	.P8....
0010	00001000 00000000 00000110 00000100 00000000 00000000 10101100 01110010	r
0018	10001001 01010000 00111000 00000100 11000000 10101000 00000001 01001001	.P8....I
0020	00000000 00000000 00000000 00000000 00000000 00000000 11000000 10101000	
0028	00000001 11110111	..

The status bar at the bottom indicates "Ethernet (eth), 14 bytes" and "Packets: 3521 · Displayed: 3521 (100.0%) | Profile: Default".

Screenshot courtesy of Wireshark.

Patch Panels

- Connection point from wall jacks (Keystone Jacks) to switch
- Punch down tool used for rear connections

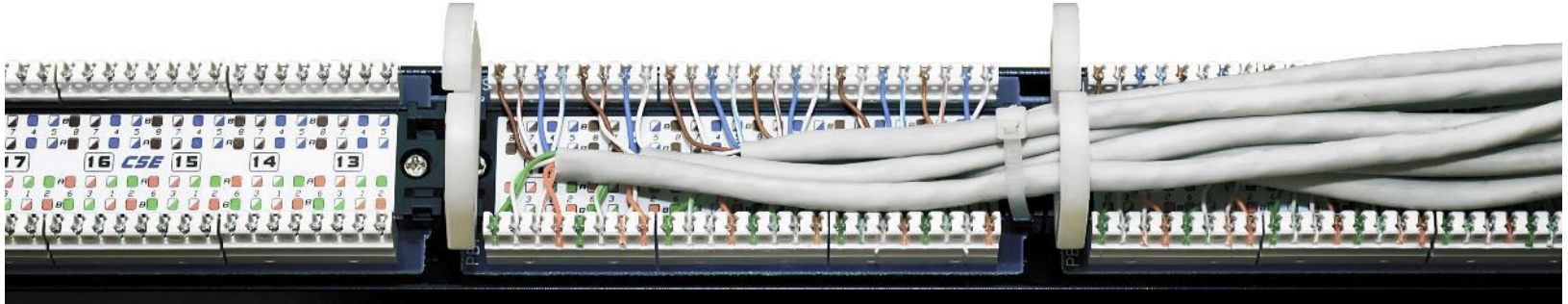


Image by plus69 © 123RF.com.

Patch Panels

- RJ-45 port used to connect to switch port on front

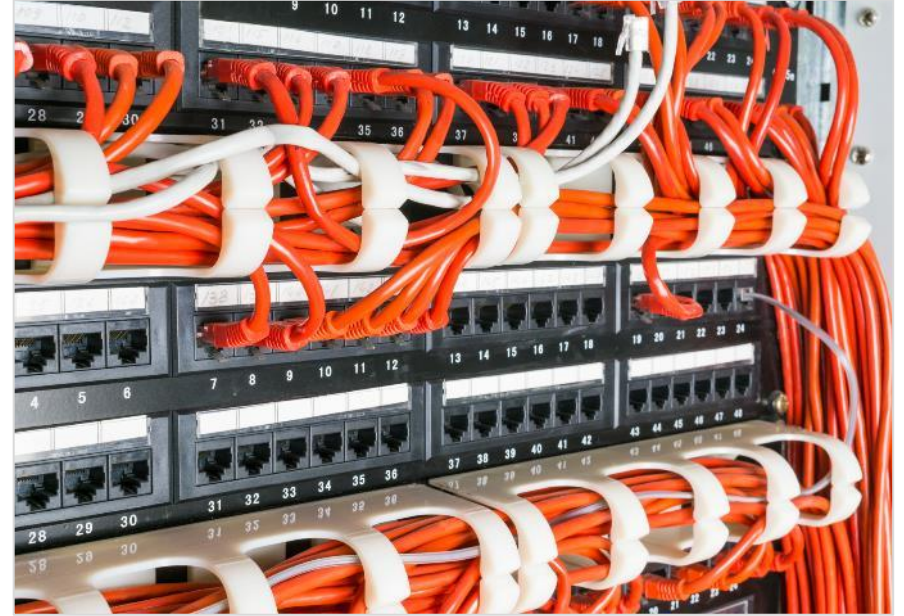


Image by Svetlana Kurochkina © 123RF.com.

Switches



Connection between end point nodes



CAM table high-speed table in switches that maps MAC addresses to specific physical switch ports



Collision domain vs. broadcast domain

Switches

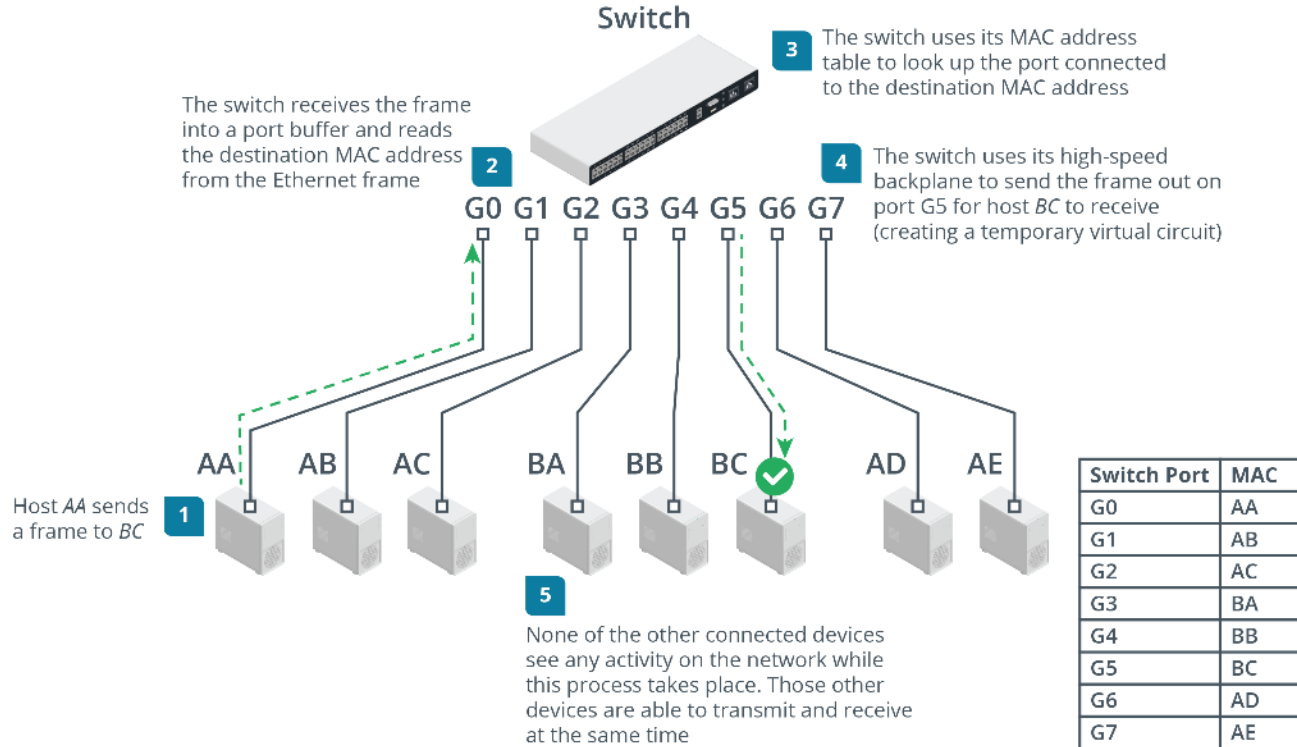


Image © 123RF.com.

Unmanaged and Managed Switches

- Unmanaged- plug and play
- Managed- management Interface for configuration

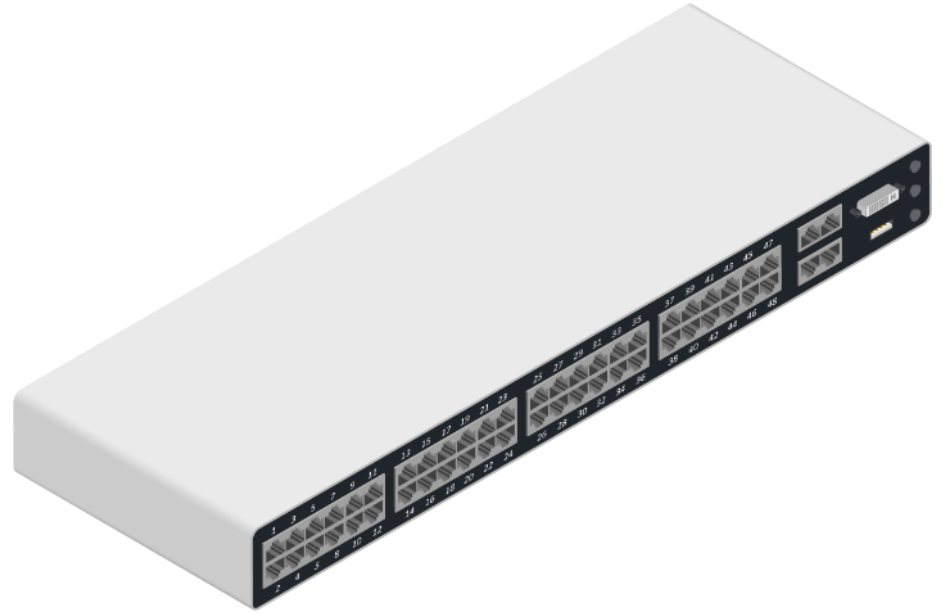


Image © 123RF.com.

Unmanaged and Managed Switches



Image © 123RF.com.

- Modular unit allow customized configuration

Unmanaged and Managed Switches

```
FastEthernet1/0/1 is up, line protocol is up (connected)
  Hardware is Fast Ethernet, address is f41f.c253.7103 (bia f41f.c253.7103)
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is 10/100BaseTX
  input flow-control is off, output flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:51, output 00:00:00, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    18 packets input, 1758 bytes, 0 no buffer
    Received 4 broadcasts (2 multicasts)
    0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 2 multicast, 0 pause input
    0 input packets with dribble condition detected
  111 packets output, 13828 bytes, 0 underruns
  0 output errors, 0 collisions, 1 interface resets
  0 unknown protocol drops
```

Power over Ethernet

IEEE standards

- 802.3af PoE or Type 1
- 802.3at PoE+ or Type 2
- 802.3bt PoE++ or Type 3 and Type 4

PoE-enabled switch

PoE injector

Lab Activity

- Lab: Select and Install a Network Adapter
(available in CertMaster Learn and CertMaster Perform)
- Lab: Connect Patch Panel Cables
(available in CertMaster Learn and CertMaster Perform)
- Lab: Connect Computers with a Switch
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Networking Hardware

Network Interface Cards



Patch Panels



Switches



Unmanaged and Managed Switches



Power over Ethernet

Lesson 5.3

Network Cable Types

Unshielded Twisted Pair

- Copper conductor
- 100m (328')

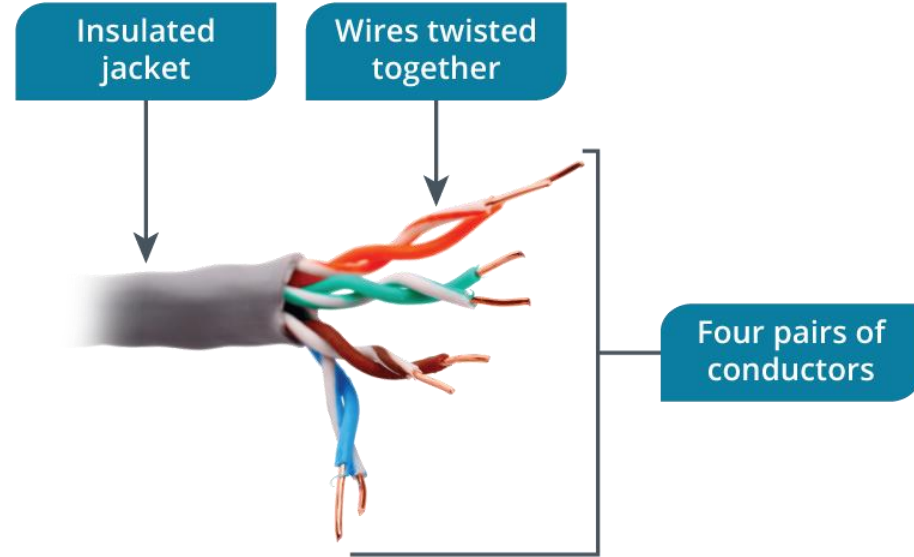


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Shielded Twisted Pair

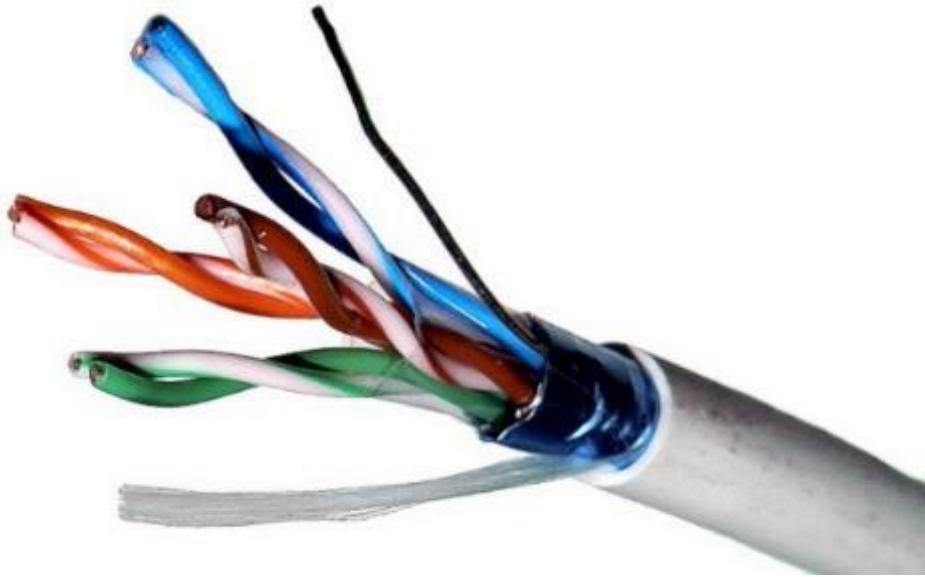


Image by Baran Ivo and released to public domain.

- Copper conductor
- Each pair has foil shielding
- May contain braided screen
- EMI protection

Cat Standards

Cat	Max. Transfer Rate	Max. Distance	Ethernet Standard Support
5	100 Mbps	100 m (328 ft)	100BASE-TX (Faster Ethernet)
5e	1 Gbps	100 m (328 ft)	1000BASE-T (Gigabit Ethernet)
6	1 Gbps	100 m (328 ft)	1000BASE-T (Gigabit Ethernet)
	10 Gbps	55 m (180 ft)	10GBASE-T (10G Ethernet)
6A	10 Gbps	100 m (328 ft)	10GBASE-T (10G Ethernet)
7	10 Gbps	100 m (328 ft)	10GBASE-T (10G Ethernet)
	100 Gbps	15 m (50 ft)	100GBASE-T (100G Ethernet)
8	25 Gbps	30 m (100 ft)	25GBASE-T (25G Ethernet)
	40 Gbps	30 m (100 ft)	40GBASE-T (40G Ethernet)

Copper Cabling Connectors

- Registered jack
- RJ-45
- RJ-11

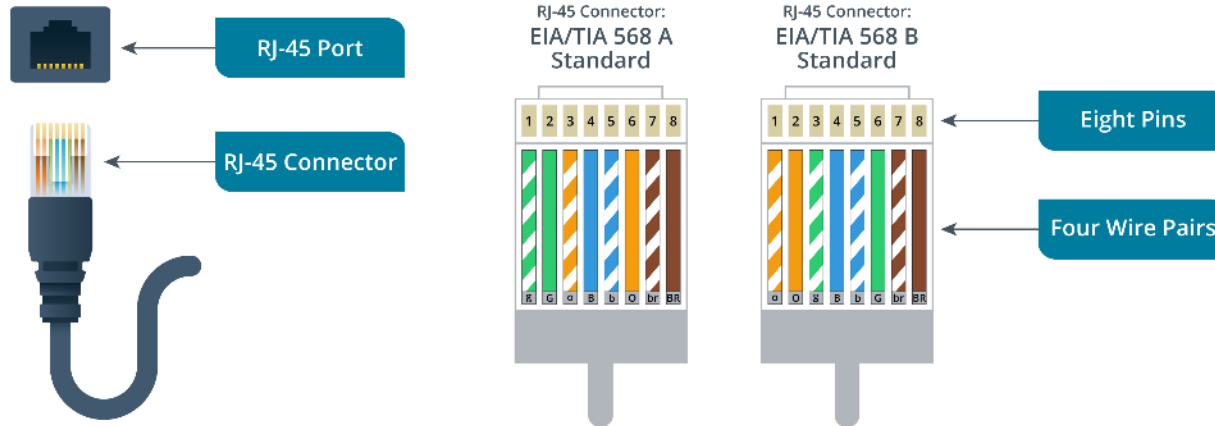


Image © 123RF.com.

Copper Cabling Installation Tools

- Cable/wire stripper
 - Used to trim back outer jacket
- Snips/cutter
 - Used to trim wire and cable to correct length
- Punchdown tool
 - Connect individual wires to an IDC or keystone jack
- Crimper
 - Attach RJ-11 or RJ-45 plug

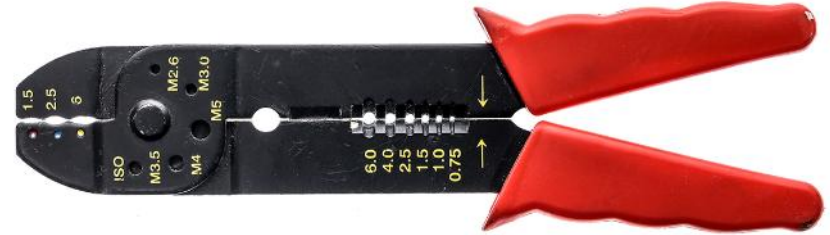


Image by gasparij © 123RF.com

Copper Cabling Installation Tools



Image by dero2084 © 123RF.com.

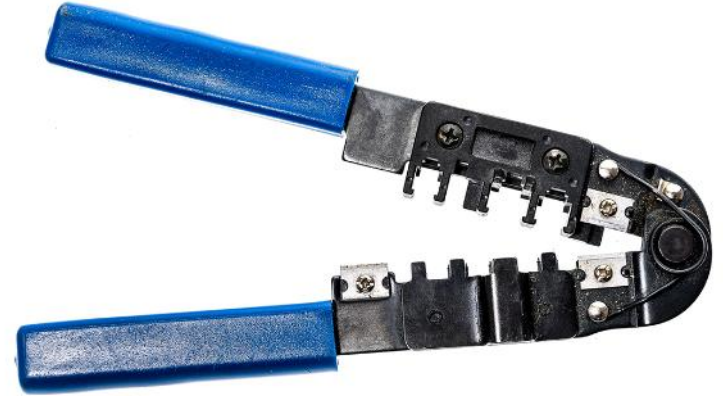


Image by gasparij © 123RF.com

Copper Cabling Test Tools

- Cable tester
 - LED indicator testing connection on individual pairs of wire
- Toner probe
 - Used to trace and locate cables in network closet or cable bundles



Image by samum © 123RF.com

Copper Cabling Test Tools

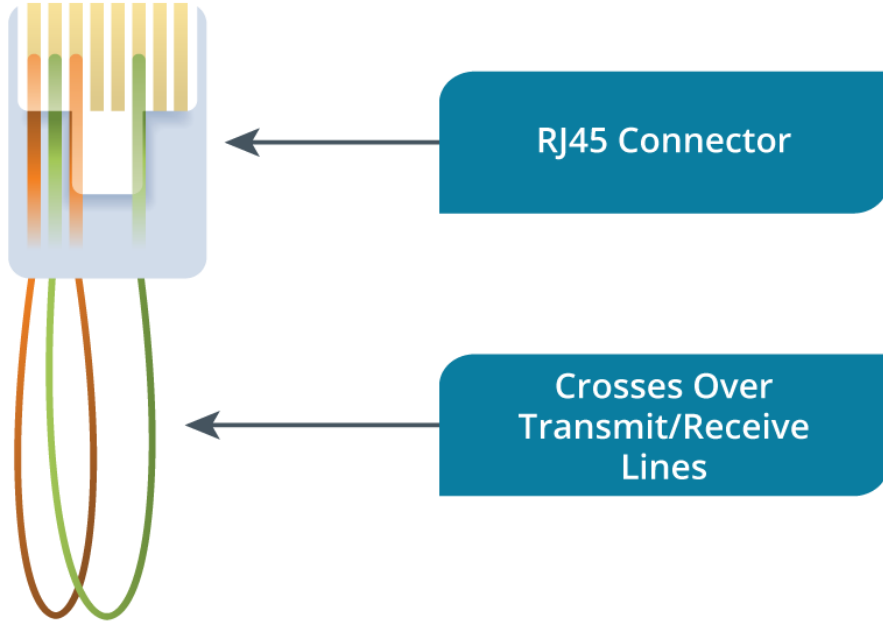


Image © 123RF.com

- Loopback plug
 - Used to test transmit and receive wire pathways

Network Taps

Passive

- Monitors traffic without interruption of dataflow by creating a copy of the data

Active

- Monitors traffic directly and then regenerates the traffic signal to continue it along its pathway to its destination

Copper Cabling Installation Considerations

Plenum
rated

Direct
burial

Optical Cabling

- Higher bandwidth
- Longer distance signaling
- Single mode vs. multimode fiber



Image by atrush © 123RF.com

Optical Cabling

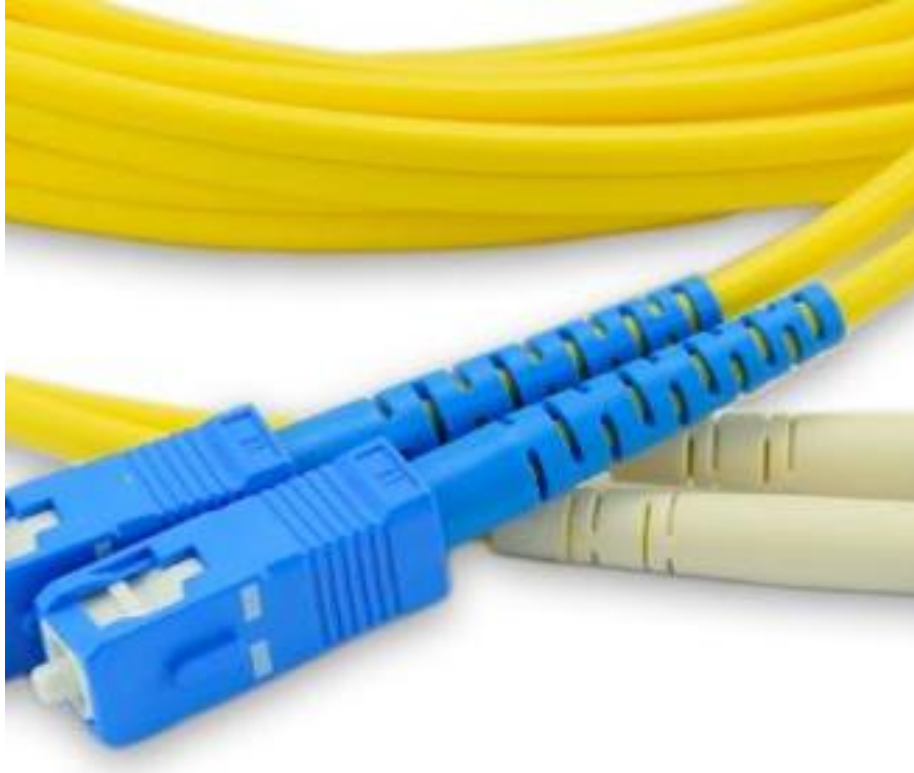


Image by YANAWUT SUNTORNKIJ © 123RF.com.

- Connectors
 - ST
 - SC
 - LC

Coaxial Cabling

- Radio Grade (RG)
 - RG58, RG59
- Connector
 - F-type
 - BNC



Image by destinacigdem © 123RF.com

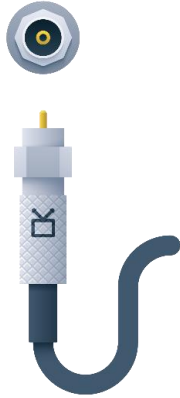


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Lesson Review: Network Cable Types

- Unshielded Twisted Pair
- Shielded Twisted Pair
- Cat Standards
- Copper Cabling Connectors
- Copper Cabling Installation Tools
- Copper Cabling Test Tools
- Network Taps
- Copper Cabling Installation Considerations
- Optical Cabling
- Coaxial Cabling

Lesson 5.4

Wireless Networking Types

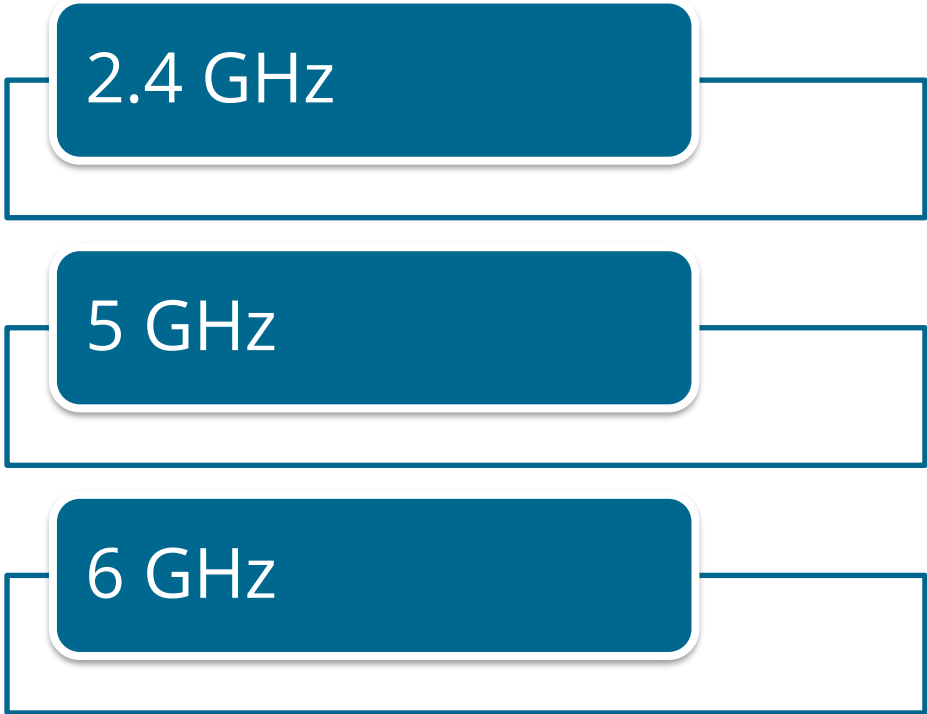
Access Points (AP)

- IEEE 802.11 standards
- BSSID- MAC of AP



Image © 123RF.com

Frequency Bands



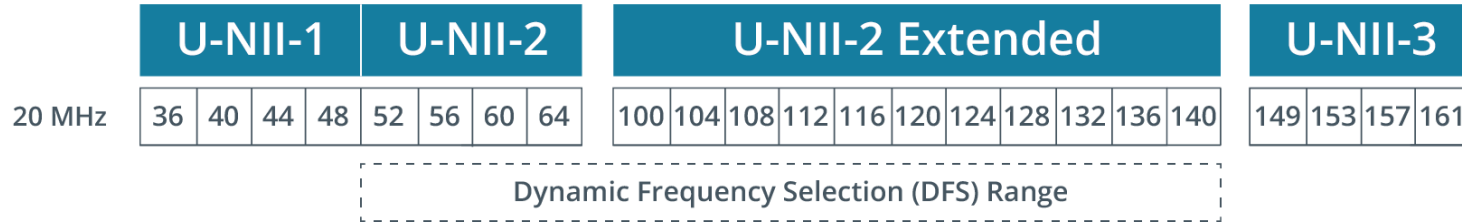
2.4 GHz

5 GHz

6 GHz

IEEE 802.11a

- 5 GHz frequency
- 54 Mbps
- DFS (Dynamic Frequency Selection)



IEEE 802.11b/g



2.4 GHz
frequency band



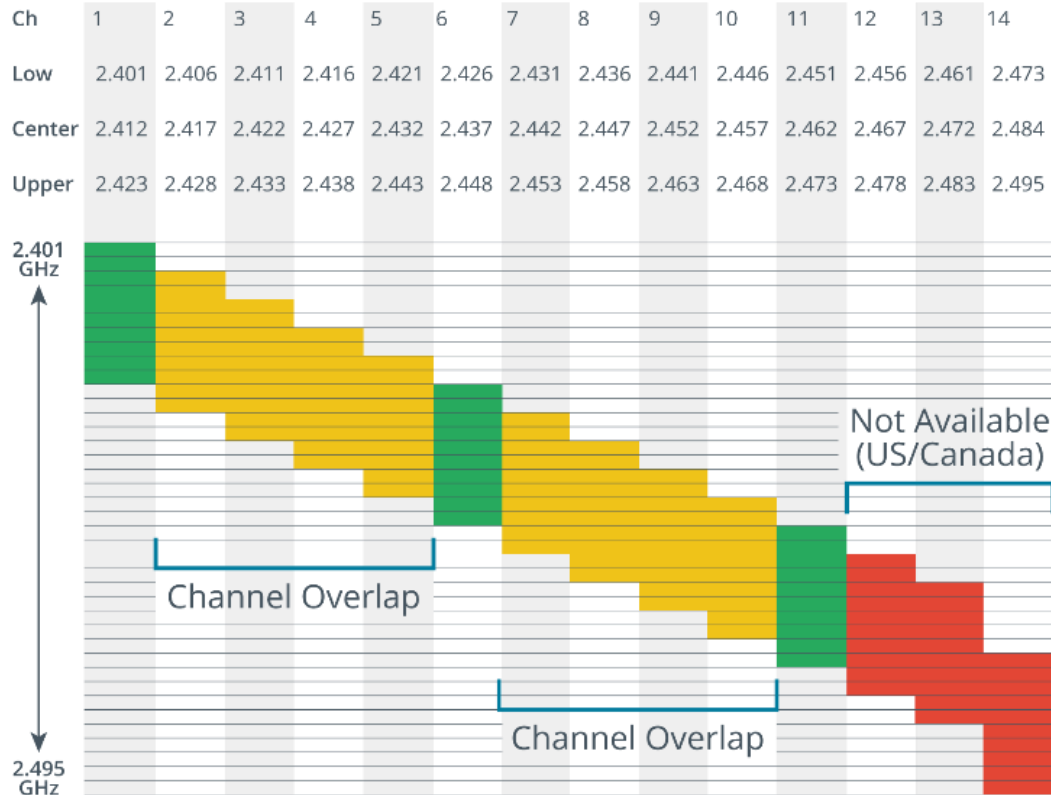
Modulation- DSSS
and OFDM



11/54 Mbps

IEEE 802.11b/g

2.4 GHz Wi-Fi Frequencies (in GHz)



802.11n

- 2.4/5 GHz dual band
- Channel bonding
- MIMO
- 600 Mbps
- Wi-Fi 4

	U-NII-1								U-NII-2								U-NII-2 Extended												U-NII-3			
20 MHz	36	40	44	48	52	56	60	64	100	104	108	112	116	120	124	128	132	136	140	149	153	157	161									
40 MHz	38		46		54		62		102		110		118		126		134			151		159										
Dynamic Frequency Selection (DFS) Range																																

Wi-Fi 5 and Wi-Fi

- 802.11ac- 5GHz only
 - MU-MIMO
 - Dual and triple band- backward compatibility with 2.4GHz systems
 - 2.1+ Gbps
- 802.11ax- 2.4/5/6GHz(6e)
 - OFDMA modulation
 - MU-MIMO
 - 1.1 Gbps 2.4GHz; 4.8 Gbps 5GHz

Wi-Fi 5 and Wi-Fi 6

	U-NII-1	U-NII-2	U-NII-2 Extended	U-NII-3
20 MHz	36 40 44 48 52 56 60 64		100 104 108 112 116 120 124 128 132 136 140	149 153 157 161
40 MHz	38 46 54 62		102 110 118 126 134	151 159
80 MHz	42 58		106 122	155
160 MHz	50		114	

Dynamic Frequency Selection (DFS) Range

Wi-Fi 7 (802.11be)



2.4/5/6 GHz



46 Gbps

Wireless LAN Installation Considerations



SSID



Frequency band



Channels

Wireless LAN Installation Considerations

TP-LINK

Archer VR900

Quick Setup

Basic

Advanced

English

Logout

Reboot

Status

Operation Mode

Network

IPTV

Wireless

Wireless Settings

WPS

MAC Filtering

Wireless Schedule

Statistics

Advanced Settings

Wireless Settings

2.4GHz | 5GHz

?

Wireless Radio:

☒ Enable

Wireless Network Name (SSID):

comptia_wlan

☐ Hide SSID

Security:

WPA/WPA2 Personal (Recommended)

Version:

☐ Auto

☒ WPA2-PSK

Encryption:

☐ Auto

☐ TKIP

☒ AES

Password:

12345670

Mode:

802.11gn mixed

Channel:

Auto

Channel Width:

Auto

Transmit Power:

☐ Low

☐ Middle

☐ High

Save

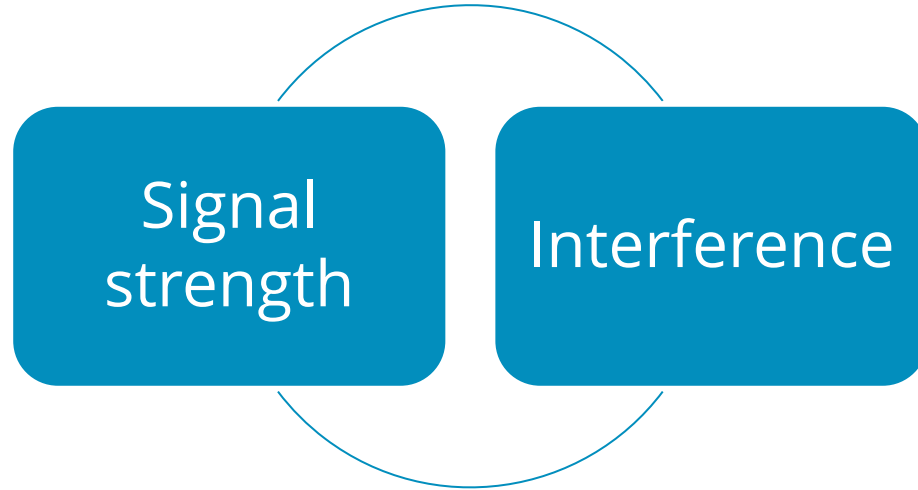
Firmware Version:0.1.0 0.9.1 v0069.0 Build 160525 Rel.38143n

Hardware Version:Archer VR900 v2 00000000

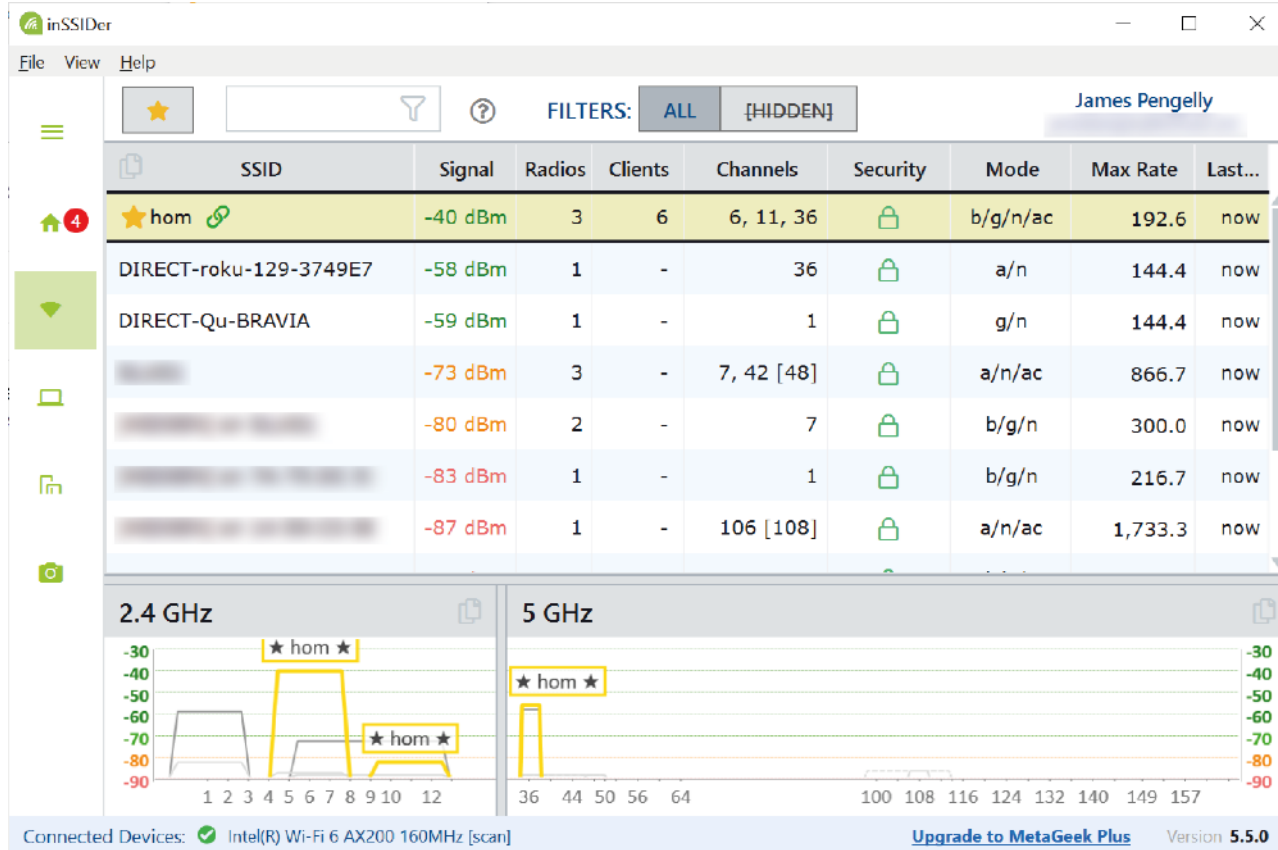
Support

Screenshot courtesy of TP-Link.

Wi-Fi Analyzers



Wi-Fi Analyzers



Long-Range Fixed Wireless

In 1985, the Federal Communications Commission (FCC) first authorized the operation of non-licensed spread spectrum systems in the 902-928 MHz, 2400-2483.5 MHz, and 5725-5850 MHzs. This decision led to technology, perhaps most notably Wi-Fi and Bluetooth, that we currently cannot imagine living without.

High-powered, long-range antennas for fixed wireless boost Wi-Fi/cellular signals for miles, using directional (dish/panel) or omnidirectional (omni) types, with parabolic antennas offering extreme point-to-point links.

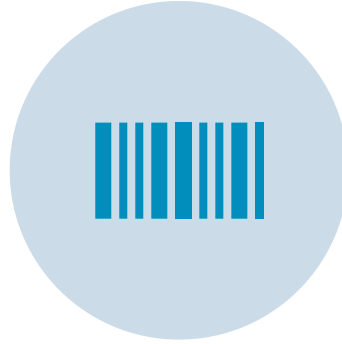
Licensed vs.
unlicensed

High-powered
antenna

Bluetooth, RFID, and NFC



BLUETOOTH PAIRING



RFID- INVENTORY TRACKING
AND BADGE ACCESS



NFC- CONTACTLESS PAYMENT
AND BADGE ACCESS USE

Bluetooth, RFID, and NFC



Image © 123RF.com

Lab Activity

- Lab: Secure a Small Wireless Network
(available in CertMaster Learn and CertMaster Perform)
- Lab: SOHO Wi-Fi Configuration Settings
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Wireless Networking Types

- Access Points
- Frequency Bands
- IEEE 802.11a
- IEEE 802.11b/g
- 802.11n
- Wi-Fi 5 and Wi-Fi 6
- Wi-Fi 7 (802.11be)
- Wireless LAN Installation Considerations
- Wi-Fi Analyzers
- Long-Range Fixed Wireless
- Bluetooth, RFID, and NFC